FI FISHINTEL GLOBAL

ENVIRONMENTAL RECONSTRUCTION REPORT

Decision-Time Environmental Reconstruction

Routine Environmental Conditions at the Stability Limit

INCIDENT WINDOW	Summer daytime operations
REPORT REFERENCE	FIG-2026-002
VESSEL CATEGORY	~14 m steel stern trawler, single-handed
REPORT VERSION	v1.0
OPERATIONAL CONTEXT	UK coastal fishing, English Channel
CLASSIFICATION	Public — Anonymised

PIPELINE VERIFIED — RECONSTRUCTION DATA POPULATED

01

Executive Summary

Incident type: Capsize and foundering. A ~14 m UK-registered steel stern trawler, operating single-handed in the English Channel, capsized and sank during recovery of a heavy trawl catch. The skipper survived after boarding the vessel's liferaft. The vessel was a total constructive loss. No fatalities.

Core question: What environmental conditions existed at the time of the capsize, and what was their relationship to the vessel's stability envelope?

Method: ERA5 atmospheric reanalysis data was extracted at the incident coordinates across a four-point time window (06:00–11:00 UTC) and processed through the FishIntel deterministic reconstruction pipeline. The reconstruction quantifies sustained wind, gust speed, significant wave height, atmospheric pressure, temperature, and visibility at each extraction point.

Key result: ERA5 reconstruction shows sustained 10 m winds of `15.3–16.7` knots (Beaufort Force 4), gusts of `22.9–23.7` knots, and significant wave height of `0.82–0.88` metres across the incident window. Wind direction was consistently northerly (`006°–010°`). Visibility was good. Atmospheric pressure was near-normal (`1013.4–1014.1` hPa). The FishIntel pipeline classifies the environmental risk as LOW to MODERATE (`36.7–41.1%`). For a vessel with adequate stability reserves, these conditions would have been routine.

Analytical finding: This was not a case of extreme weather overwhelming a vessel. The environmental conditions were moderate, stable, and foreseeable — they matched the published forecast. The evidence indicates that the capsize occurred when the vessel's stability reserves were critically thin. Stability analysis conducted during the subsequent investigation placed the vessel only 35 mm above the danger-of-capsize zone, where the maximum recommended sea state is 0.8 m. ERA5 shows significant wave height of 0.82–0.88 m, which was below the amber-zone maximum of 1.6 m but close to, and marginally above, the 0.8 m threshold associated with the red danger-of-

capsize zone. Routine environmental conditions became unsafe because vessel-specific safety margins were already critically reduced through accumulated modifications, low fuel, seized freeing ports, and operational loading from a high gantry point. The information required to calculate this vulnerability existed before the vessel sailed, but no such calculation had been undertaken at the time of the incident.

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Incident Overview

Vessel Profile

TYPE	Decked stern trawler
FLAG	United Kingdom
LENGTH OVERALL	~14 m
GROSS TONNAGE	~19
YEAR OF BUILD	Late 1980s
CONSTRUCTION	Steel
PROPULSION	Single screw, blade rudder
MANNING AT INCIDENT	1 (single-handed)
NORMAL MANNING	3 (skipper + 2 crew)
FREEBOARD (MEASURED)	305 mm
STABILITY ASSESSMENT	None conducted prior to incident
FUEL ON BOARD	~9% of capacity

The vessel departed its home port at approximately 0640 local time, heading to established fishing grounds in the English Channel to trawl for demersal fish species. The skipper was operating single-handed for the second consecutive day due to crewing difficulties. The vessel had been unable to take fuel before departure as the harbour fuel pump was not working.

The vessel arrived at the fishing grounds approximately one hour later and deployed a single demersal trawl net. After approximately two hours of

trawling, the skipper began hauling the first catch. The powered net drum struggled under an excessive load. The skipper varied course and speed to wash accumulated mud and sand from the net. The first Gilson rope parted under load — an early indicator of excessive weight in the cod end.

After rigging a replacement Gilson rope over the starboard side of the aft gantry, the skipper recovered a partial load of approximately 1.5 tonnes of mud, sand, gravel, starfish and some fish onto the main deck. At least another 1.5 tonnes remained in the net. When the cod end cleared the sea surface during a final recovery attempt, it slewed to starboard. The vessel heeled, the deck edge became immersed, water poured on deck, the catch already on deck slid to starboard, and the vessel capsized rapidly to starboard and sank.

The skipper, wearing a PFD with a personal locator beacon, entered the water, swam to the vessel's liferaft, and raised the alarm via mobile phone. The skipper was recovered uninjured approximately 40 minutes later by a nearby vessel.

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Environmental Conditions Timeline

ERA5 reanalysis data was extracted at the incident coordinates across four timestamps spanning the operational window from departure through capsize. All values are produced by the FishIntel deterministic pipeline from ECMWF ERA5 source data.

TIME (UTC)	EVENT	WIND (KT)	GUSTS (KT)	SWH (M)	RISK
06:00	*Departure*	15.3	23.1	0.82	MODERATE (40.0%)
08:00	*Fishing operations*	16.3	23.7	0.88	MODERATE (41.1%)
10:00	*Hauling / catch recovery*	15.5	23.0	0.87	LOW (37.6%)
11:00	*Capsize*	16.7	22.9	0.85	LOW (36.7%)

INTERPRETATION

Environmental conditions were stable across the entire five-hour operating window. Sustained wind varied by only 1.4 knots (15.3–16.7 kt). Gusts varied by 0.8 knots (22.9–23.7 kt). Significant wave height varied by only 0.06 m (0.82–0.88 m). There was no sudden deterioration, no squall, and no environmental step-change. The conditions at departure were substantively the same as the conditions at capsize.

The FishIntel risk classification marginally decreased from MODERATE to LOW across the window, reflecting a slight moderation consistent with the investigation narrative of wind dropping from Force 6 to Force 4–5 through the morning. The variable that changed during the operating window was not the environment — it was the vessel's effective stability, progressively reduced by operational actions.

04

Narrative Weather Description

The following environmental conditions were described in the original accident investigation report. These represent the qualitative and semi-quantitative descriptions available in the published account.

PARAMETER	NARRATIVE DESCRIPTION	SOURCE & NOTES
Wind (morning)	Northerly, Force 6 22–27 knots	*Early-day conditions.*
Wind (at capsize)	Northerly, Force 4 to 5 11–21 knots	*Accident particulars. Dropping to Force 3 by end of day.*
Seas (Force 6)	2.5 m	*Associated with morning Force 6 winds.*
Seas (Force 4)	1.5 m	*"Reducing to 1.5 m in the force 4 winds."*
Water temperature	16°C	—
Visibility	Good, occasionally moderate	—
Forecast accuracy	Conditions matched Met Office forecast	*Skipper consulted forecast before departure.*

Analytical note: The investigation provides two wind descriptions at different points: Force 6 for morning conditions and Force 4–5 for the time of capsize. These are consistent with a moderating wind through the morning. The sea-state

description of 1.5 m in Force 4 conditions is compared against ERA5 significant wave height in the reconstruction.

05

Environmental Reconstruction

The following table compares the environmental conditions described in the narrative account with the values reconstructed by FishIntel using ECMWF ERA5 atmospheric reanalysis data at the incident position. The 11:00 UTC extraction (nearest hour to capsize) is used as the primary comparison point.

PARAMETER	NARRATIVE	ERA5 RECONSTRUCTION	ASSESSMENT
Wind speed (sustained)	Force 4–5 (11–21 kt)	16.7 kt	***Consistent.*** *ERA5 confirms Beaufort Force 4 at capsize time.*
Wind gusts	*Not stated*	22.9 kt	*Gust factor 1.37. Intermittent gusts reached Force 6 equivalent.*
Wind direction	Northerly	010°	***Consistent.*** *Confirmed nearly due north throughout.*
Significant wave height	1.5 m (in Force 4)	0.85 m	***Divergent.*** *ERA5 gridded SWH lower than narrative description. See analytical note.*

PARAMETER	NARRATIVE	ERA5 RECONSTRUCTION	ASSESSMENT
Wave period	***Not stated***	**4.0 s**	*Short-period wind waves consistent with local Force 4 generation.*
Atmospheric pressure	***Not stated***	**1013.7 hPa**	*Near-normal. No significant synoptic forcing.*
Air temperature	***Not stated*** (water 16°C)	**16.1°C**	***Consistent*** *with stated water temperature of 16°C.*
Visibility	Good, occasionally moderate	**Good** (dewpoint spread 4.4°C)	***Consistent.***
Wind shear (10–100 m)	***Not stated***	**4.5 kt**	*Moderate shear. Wind at gantry height close to 10 m reference.*

16.7 kt SUSTAINED WIND (10 M)	**22.9** kt GUST SPEED	**0.85** m SIG. WAVE HEIGHT	**1013.7** hPa PRESSURE MSL

ANALYTICAL NOTE — WAVE HEIGHT DIVERGENCE

The investigation report describes seas of 1.5 m in Force 4 conditions. ERA5 reconstruction shows a significant wave height of 0.85 m at the incident coordinates. This divergence of approximately 0.65 m requires transparent treatment.

ERA5 is a gridded reanalysis product at approximately 31 km horizontal resolution. It does not capture localised effects including: shallow-water wave steepening over the variable-depth seabed; tidal current interaction with northerly waves; or the visual sea-state impression experienced at deck level on a ~14 m vessel, which includes whitecapping, spray, and vessel motion that compounds the perceived severity beyond the statistical significant wave height.

This report presents both values. The ERA5 value of 0.85 m represents the gridded significant wave height. The narrative value of 1.5 m represents the local description from the investigation. The vessel stability comparison considers both as the lower and upper bounds of the actual sea state.

Unit Conversion Chain — Wind Speed

ERA5 source: **8.5918** m/s × 1.94384 = **16.7 kt**

Unit Conversion Chain — Pressure

ERA5 source: **101366.25** Pa ÷ 100 = **1013.7 hPa**

Unit Conversion Chain — Temperature

ERA5 source: **289.28** K − 273.15 = **16.1°C**

06

Decision-Time Environment

The following chronology presents the sequence of events from departure through capsize, with the reconstructed environmental state at the nearest ERA5 extraction point. Decision gates — points where the operator had the opportunity to alter the outcome — are identified.

0540 UTC — Departure Decision

Vessel departs home port, single-handed

Heading NE toward fishing grounds. Unable to take fuel (pump not working). Fuel on board approximately 9% of capacity. Skipper had consulted the Met Office forecast. Second consecutive day operating alone.

```
ERA5 06:00: 15.3 kt N (006°) | Gusts 23.1 kt | SWH 0.82 m | 1014.1 hPa
| MODERATE (40.0%)
```

~0700 UTC

Fishing gear deployed

Single demersal trawl net deployed at the fishing grounds.

```
ERA5 08:00: 16.3 kt N (007°) | Gusts 23.7 kt | SWH 0.88 m | 1014.1 hPa
| MODERATE (41.1%)
```

~0930 UTC — Hauling Decision

Starts hauling — heavy load evident

Net drum struggling under excessive load. Skipper varies course and speed to wash out debris. First Gilson rope parts under load — indicator of excessive weight.

```
ERA5 10:00: 15.5 kt N (007°) | Gusts 23.0 kt | SWH 0.87 m | 1013.4 hPa
| LOW (37.6%)
```

~1020 UTC

Partial load recovered to deck

~1.5 t of mud, sand, starfish and fish released onto main deck. Replacement Gilson rope rigged on aft gantry. At least 1.5 t remains in net.

~1031 UTC — Final Recovery Decision

Final cod end recovery attempt

Vessel on south-westerly course, wind and sea on starboard quarter. Main load on Gilson rope over the starboard side of the aft gantry. Operating single-handed: operator simultaneously managing heading, winch, and catch.

~1050 UTC

Cod end clears water — slews to starboard

Full weight of suspended catch acts from the highest point on the vessel. Vessel heels to starboard. Deck edge immersed. Water on deck. Catch slides to starboard. Capsizing moment increases continuously. No means to control transverse slewing.

1056 UTC — CAPSIZE

Vessel capsizes to starboard and founders

Continuous roll to starboard. Downflooding through open watertight doors. Vessel sinks bow-first. Skipper enters water, swims to liferaft.

```
ERA5 11:00: 16.7 kt N (010°) | Gusts 22.9 kt | SWH 0.85 m | 1013.7 hPa
| LOW (36.7%)
```

~1140 UTC

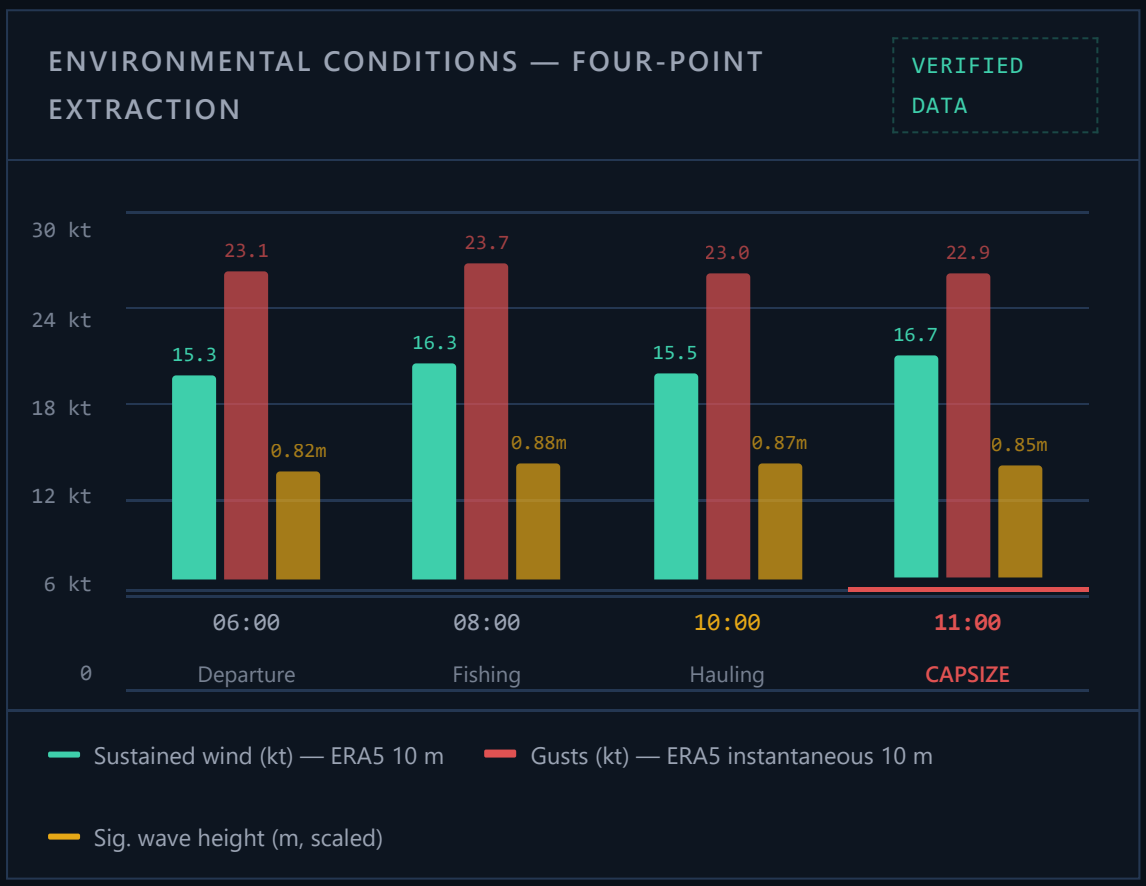
Skipper recovered from liferaft

Skipper uninjured. Recovered by a nearby vessel responding to coastguard broadcast.

07

Environmental Profile — Verified Data

The following graphic presents the verified four-point environmental profile across the incident window. Each data point is derived from the FishIntel reconstruction pipeline at the coordinates and timestamp shown.



INTERPRETATION

The four-point extraction reveals a flat environmental profile. Unlike the first FishIntel flagship reconstruction — which showed a slowly escalating storm regime — this case shows environmental conditions that were functionally constant across the entire operating window. All four extraction points show sustained wind within a 1.4 kt range and wave height within a 0.06 m range. The visual uniformity of the chart is itself the finding: the environment did not change. The vessel's stability did.

08

Decision Window / T_{irr} Assessment

FishIntel's Time-to-Irreversibility (T_{irr}) framework identifies when an operational situation becomes irreversible — the point beyond which no reasonable intervention by the vessel's crew can change the outcome. In this case, the T_{irr} analysis addresses a fundamentally different mechanism from the first flagship reconstruction: the decision window closed not through environmental escalation, but through progressive consumption of stability reserves.

Decision Window Findings

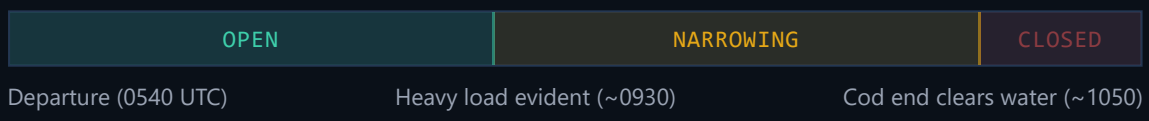
ENVIRONMENTAL RISK REGIME	LOW–MODERATE throughout (36.7–41.1%)
ENVIRONMENTAL ESCALATION	None. Conditions stable across incident window.
STABILITY REGIME AT DEPARTURE	Amber zone (Wolfson Guidance) — low level of safety
STABILITY MARGIN TO RED ZONE	35 mm freeboard
CRITICAL DECISION GATE	Final cod end recovery (~1031 UTC) with ~3 t total load, single-handed
POINT OF IRREVERSIBILITY	Cod end clears water and slews to starboard (~1050 UTC)

T_{irr} in this case is driven by vessel dynamics, not environmental progression. The environmental state remained within the same regime throughout. Irreversibility was

reached when the suspended catch created a capsizing moment that exceeded the vessel's remaining righting capacity.

The T_{irr} framework classifies operational situations into three regimes:

REGIME	CLASSIFICATION	MEANING
Regime 1	**Reversible**	*Conditions permit effective intervention. The decision window is open. Corrective action remains available and viable.*
Regime 2	**Conditionally reversible**	*Intervention is possible but constrained. The window is narrowing. Success depends on immediate action and favourable conditions.*
Regime 3	**Irreversible**	*No reasonable intervention can alter the outcome. The situation has passed T_{irr}. The operational decision that could have changed the outcome had to be taken earlier.*



At departure, the situation was in Regime 1: the decision window was fully open. The environment was moderate and foreseeable. When the excessive load became apparent and the first Gilson rope parted, the situation moved into Regime 2: corrective action was still available (the catch could have been released), but the margin was narrowing with each additional tonne on deck. When the cod end cleared the water and slewed to starboard, the situation entered Regime 3 within seconds: the capsizing moment exceeded the remaining righting capacity and no intervention could arrest the roll.

The environmental conditions played no escalating role in this progression. The T_{irr} transition from Regime 1 to Regime 3 appears to have been determined by the operational loading sequence acting against a vessel with critically depleted stability reserves.

09

Risk Analysis

36–41

Environmental Risk:

LOW–MODERATE

Pipeline classification across all four extraction timestamps

The FishIntel risk assessment integrates multiple environmental drivers to produce a single quantified score. The drivers for this case include:

RISK DRIVER	VALUE	CONTRIBUTION
Sustained wind	16.7 kt	*Beaufort Force 4. Within normal operating parameters.*
Gust speed	22.9 kt	*Intermittent Force 6 equivalent. Gust factor 1.37.*
Significant wave height	0.85 m	*Below the amber-zone maximum (1.6 m) but close to, and marginally above, the 0.8 m red danger-of-capsize threshold for this vessel's stability condition.*
Wind vs tide	with	*Wind and waves aligned. No opposing-tide amplification.*
Visibility	good	*No impairment to situational awareness.*
Wind shear	4.5 kt	*Moderate. Wind at gantry height close to surface reference.*

The environmental risk classification of LOW–MODERATE is correct for a vessel with adequate stability reserves. The analytical significance of this case lies in

the compound interaction between moderate environmental forcing and vessel-specific vulnerability factors: freeboard only 35 mm above the Wolfson danger zone; fuel at 9% capacity (raising VCG); three freeing ports seized and two covered (trapping deck water); approximately 3 tonnes of catch loaded from a high gantry point; and single-handed operation limiting the operator's capacity to manage multiple simultaneous tasks.

◆ FORESEEABILITY TEST

This report assesses foreseeability by applying the following test to the reconstructed environmental evidence:

- The environmental data required to assess the risk existed and was accessible at the time of the decision (ERA5 reanalysis confirms this retrospectively; Met Office forecast data confirms it prospectively).
- The vessel-specific data required to calculate stability limits existed: the Wolfson Guidance method requires only length overall and beam, both known quantities. This calculation was never performed.
- Had the Wolfson method been applied, it would have identified that the vessel was operating in the amber zone with a maximum recommended sea state of 1.6 m and only 35 mm of freeboard margin above the red zone (max 0.8 m). ERA5 shows SWH of 0.82–0.88 m, which was below the amber-zone maximum but close to, and marginally above, the 0.8 m red-zone threshold.
- The environmental conditions were foreseeable and matched the published forecast. There was no sudden, unforeseeable step-change. The conditions were continuous and stable throughout the operating window.

This report does not attribute fault or negligence to any individual. The foreseeability test evaluates the environmental evidence and vessel stability data, not the decisions of specific persons. Its purpose is to establish whether the conditions that contributed to the incident were objectively knowable before the outcome occurred.

Key Analytical Finding

KEY FINDING

Environmental reconstruction confirms that this vessel capsized in conditions the FishIntel pipeline classifies as LOW to MODERATE risk: sustained wind 15.3–16.7 knots (Beaufort Force 4), significant wave height 0.82–0.88 metres, good visibility, and stable atmospheric pressure across the entire operating window. These conditions were moderate, foreseeable, and matched the published forecast. For a vessel with adequate stability reserves, they would have been unremarkable. The evidence indicates that the capsize arose from the interaction of routine environmental conditions with a vessel whose stability reserves had been critically reduced through accumulated modifications, low fuel state, seized freeing ports, and the progressive operational loading of a heavy catch from a high gantry point. The information required to calculate the vessel's vulnerability to these conditions existed before departure — the Wolfson Guidance method requires only length and beam — but no such calculation had been undertaken at the time of the incident.

PIPELINE CLASSIFICATION

Environmental risk LOW–MODERATE (36.7–41.1%) across all extraction timestamps. Conditions stable: wind 15.3–16.7 kt, gusts 22.9–23.7 kt, SWH 0.82–0.88 m. No environmental escalation detected. Vessel stability condition placed it close to, and marginally above, the Wolfson 0.8 m red danger-of-capsize threshold (ERA5 SWH 0.85 m, margin 35 mm freeboard). Analysis indicates the capsize was

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associated with compound operational loading against critically
depleted stability reserves, not with environmental exceedance.
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This finding is derived from the quantified environmental reconstruction and the vessel stability data from the investigation, not from the narrative account. It represents what was objectively knowable at the time the critical operational decision was made, based on environmental data and vessel parameters that existed and were accessible.

This report does not attribute fault or negligence to any individual. The operator was experienced, consulted the forecast, wore a PFD, and survived the incident. The purpose of the reconstruction is to quantify the environmental conditions and their relationship to the vessel's stability state.

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Evidence Integrity

This report is a rendering of outputs produced by the FishIntel deterministic reconstruction pipeline. All environmental values presented are traceable to their source data through an auditable processing chain.

PIPELINE VERIFICATION

DATA SOURCE	ECMWF ERA5 Atmospheric Reanalysis
PROCESSOR VERSION	5.3.0
RUNNER VERSION	2.1.0
RAW FORMAT	era5_raw_v1
RECONSTRUCTION TIMESTAMP	2026-03-31T21:45:00Z
EXTRACTION TIMESTAMPS	06:00, 08:00, 10:00, 11:00 UTC
CASE POSITION	50.52°N, 3.27°W
CASE DATE	2021-06-18

The pipeline operates as follows: ERA5 NetCDF data is extracted at the incident coordinates and time window by the ERA5 processor, producing raw values in source-native units (m/s, Pa, K, m). These raw values are converted using explicit arithmetic (m/s × 1.94384 = knots; Pa ÷ 100 = hPa; K − 273.15 = °C). The reconstruction is deterministic: the same inputs will always produce the same outputs.

No manual adjustments, interpolations, or subjective corrections are applied at any stage.

Methodology

FishIntel Global reconstructs the environmental conditions surrounding maritime incidents using ECMWF ERA5 atmospheric reanalysis data — the same dataset relied upon by meteorological agencies, climate researchers, and aviation safety authorities worldwide.

ERA5 provides hourly estimates of atmospheric, land-surface, and ocean-wave variables on a global grid at approximately 31 km horizontal resolution. It assimilates millions of observations per cycle and is continuously quality-controlled by the European Centre for Medium-Range Weather Forecasts. ERA5 data is freely available and independently verifiable by any party.

The FishIntel reconstruction pipeline extracts environmental variables at the precise coordinates and time window of each incident. All values pass through a unit-gated validation layer that enforces explicit conversion factors and plausibility bounds using Decimal arithmetic — preventing the silent unit errors that are a known source of analytical failure in environmental data processing. The pipeline is deterministic: identical inputs always produce identical outputs.

This reconstruction operates within defined boundaries. ERA5 is a gridded reanalysis product, not a point measurement. It does not capture localised effects smaller than its grid resolution, such as harbour-scale wind funnelling or shallow-water wave refraction. Where a case involves such effects, the reconstruction states this limitation explicitly. The system is designed to produce defensible evidence, not to overstate its precision.

FishIntel’s role is complementary to national maritime accident investigation. Investigation bodies establish causes and circumstances. FishIntel quantifies the environmental context in which operational decisions were made, using

data sources that existed at the time of those decisions. The focus is on what was knowable, not on who is to blame.

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Closing Statement

Maritime incidents are frequently described after the fact using qualitative or semi-quantitative environmental descriptions: “strong winds,” “heavy swell,” “poor visibility.” These descriptions are valuable for narrative understanding but do not quantify the decision-time environment with the precision required by insurers, regulators, and legal professionals.

This reconstruction addresses a different and equally important category of incident: one where the environmental conditions were not extreme. Force 4 winds. Sub-metre wave height. Good visibility. The kind of conditions in which experienced fishermen go to work. Environmental reconstruction confirms this ordinariness — and reveals why it matters.

The vessel that capsized on this day had a stability margin so thin that conditions which appeared routine were, for this specific vessel in this specific configuration, at the boundary of the danger zone. The Wolfson Guidance — a calculation requiring only two measurements, both known — would have shown this. The Met Office forecast was accurate. The environmental conditions were foreseeable. What was missing was the bridge between environmental knowledge and vessel-specific vulnerability.

This is the gap that foreseeability intelligence is designed to close. Not by predicting accidents, but by quantifying the conditions under which they become possible — before the critical moment arrives.

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